AP PRECALCULUS

Unit 1 • Lesson 1.8 practice

Lesson 1.8 • Rational Functions and Zeros

4 multiple-choice questions and 1 free-response question

Question 1

What are the zeros of $r(x) = \dfrac{x^2 - 9}{x + 2}$?

(A) -2

(B) 3 and -3

(C) 9

(D) 3 only

Question 2

What are the zeros of $r(x) = \dfrac{x^2 - 4x}{x - 4}$?

(A) 0 and 4

(B) 4 only

(C) 0 only

(D) There are none.

Question 3

What are the zeros of $r(x) = \dfrac{(x + 1)(x - 5)}{(x - 5)(x + 3)}$?

(A) -1 only

(B) -1 and 5

(C) 5 only

(D) -3 and -1

Question 4

What is the solution set of $\dfrac{x + 2}{x - 1} < 0$?

(A) $(-\infty, -2)$

(B) $(1, \infty)$

(C) $(-\infty, -2) \cup (1, \infty)$

(D) $(-2, 1)$

Question 5 (Free response)

Let $r(x) = \dfrac{x^2 - x - 2}{x^2 - 9}$.

(a) Factor the numerator and denominator, and find the zeros of r.

(b) Make a sign chart for r and solve $r(x) > 0$.